



Consumer Acceptance of Product as a Service Leasing Model for Home Appliances Using an Extended Theory of Planned Behavior in Ecuador

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Abstract

The home appliances sector's role in the depletion of critical raw materials and electrical and electronic waste generation highlights the need for different circular business models. Use-oriented product-as-a-service (PaaS) leasing model offers a great opportunity for original equipment manufacturers (OEMs) to provide customers access to different appliances while guarding the ownership, facilitating to close the supply chain. However, one major obstacle in the adoption of PaaS leasing models is consumers lack of trust. This is particularly true in developing countries such as Ecuador, where the culture of ownership is well attached in their consciousness. Supported in the theory of planned behavior (TPB) through the incorporation of subjective norms, perceived behavioral control, and attitude towards behavior, driving the intentions to behave ecologically, this research presents an extended framework by incorporating two additional constructs, environmental consciousness and external influences in moderation and mediation roles, respectively, to predict consumers' leasing intentions. By using the data from 1,026 home owners in Ecuador and through partial least square structural equation model (PLS-SEM), the findings reveal that attitude towards behavior ($\beta=0.492$, $p<0.001$) and subjective norms ($\beta=0.347$, $p<0.001$) are the most important drivers of intentions to behave ecologically, explaining almost 68% of its variance. The high explained variance of consumers' leasing intentions (65%) supports the important role of intentions to behave ecologically bridging the intention-behavior gap. Environmental consciousness acts as a powerful moderator showing that the intention-behavior gap is lower for high conscious individuals. External influences play a partial mediation between intentions and behaviors, fortifying but not being fully accounted for their relationship. Policymakers and OEMs should focus on advertising the environmental benefits of the model and the intrinsic benefits for the individuals adopting them (energy savings, cost reduction, access to high-performant appliances) providing incentives to enhance positive attitudes and peer acceptance to encourage the eco-friendly leasing option.

Keywords Leasing Intentions · Home Appliances · Theory of Planned Behavior · Environmental Consciousness · External Influences · Structural Equation Modelling

Introduction

It is widely recognized that nowadays' production and consumption practices are major factors in global environment-related issues. This is particularly true for the electrical and electronic equipment sector (EEE), that constantly grows in popularity and utilization [7], while simultaneously, generating a rising accumulation of used EEE driven by the shorter lifespan of electronic devices, and finally becoming a waste problem having accounted more than 60 million tons in 2022 and expected to increase to 82 by 2030 [12]. The home appliances sector (e.g., dishwashers, fridges, washing machines) exert environmental pressure throughout their entire lifecycle, consuming approximately 0.6 million tons of metals (steel, aluminum, copper, and other critical and rare elements) and 0.2 million tons of plastics annually during their manufacture, while only 35% is collected for recycling purposes within the EU yearly [17]. This industry represents a significant opportunity for waste reduction purposes through the integration of modern circular business models, such as product-as-a-service (PaaS), which has been hailed as a crucial tactic to isolate the consumption of various resources and reduce waste generation with the value modeling, as noted by Tukker [83].

While in traditional purchasing models (selling business models), consumers buy and own products, typically from an intermediary (retailer), bearing full responsibility for their maintenance, operation, and disposal [53], a PaaS model implies a long-term relationship between an original equipment manufacturer (OEM) and customers, amalgamating the offer of different products and including specific services within a PaaS contract [45, 85]. PaaS solutions have the potential to contribute to resource efficiency and circularity through the provision of operational support to customers, product maintenance activities, product sharing schemes, and delivery of functional results [66]. They can also promote OEMs willingness to redesign their products for longevity, reuse, repair and recycling, reducing waste generation [35, 41].

Researchers typically categorize PaaS offerings using a threefold classification: oriented to the product, oriented to the use, and oriented to the results [11, 60, 83]. Product-oriented PaaS models offer a physical and tangible product in the conventional sales model, transferring its property to the consumer, but also including additional extra services through a fee payment, ensuring the owners their products proper operation and longevity [44]. Use-oriented PaaS models, suppliers offer the accessibility of customers to their desired product, however, property is not transferred, being the responsibility of the supplier to keep the product working during the use-phase and to recuperate the product at the end of use [44]. With result-oriented PaaS models, instead of offering the product to the customer transferring or not the property to them, the focus here is delivering the outcome, making customers pay for the utility delivered by the product [64].

Among these, use-oriented PaaS is particularly notable for its environmental benefits, since it facilitates the cycling of products among multiple end-users, integrates take-back and end-of-life management strategies, intensifies product use, and prolongs product lifespans. With use-oriented PaaS, customers can gain access to responsibility free high-quality products and services by leasing or renting [35, 48]. OEMs' income enters through different fees (weekly, monthly, yearly, or other accepted by the stakeholders) paid by the customers for the time of the leasing contract, with a final objective of getting multiple contracts per customer. Contract duration can be proactively managed to capture optimal product value and minimize repair, refurbishment and remanufacture costs [36]. This approach reduces

the frequency of product replacements and minimizes landfill waste, fostering a more sustainable product lifecycle [13].

As of this day, the practical implementation of PaaS offerings remains limited, being predominantly dominated by leasing offers from retailers. However, retailers lack the capacity of closing the supply chain and are most likely to resell the products in second-hand markets, once they reach the maximum number of leasing cycles, entering once again in the traditional lineal economy. OEMs' adoption and execution of PaaS leasing models can help closing the material loop offering different value retention processes such as reuse, repair, and refurbish [41], however, their implementation face numerous challenges, primarily stemming from economic, cultural, and operational barriers. Many customers are hesitant to embrace subscription-based or usage-based models, primarily due to privacy data concerns, the ownership culture of the population, consumers tendency to frequently upgrade their products, and the relatively newness of PaaS models, particularly in emerging markets [44, 48, 66]. With the study of behavioral patterns, researchers, policymakers, and practitioners can understand the key factors that shape consumers intentions to adopt home appliances leasing.

The theory of planned behavior (TPB) offers a well-tested theoretical basis to analyze individual behavior. It comprises the interplay of social and cognitive characteristics through constructs such as subjective norms, attitude towards behavior, and perceived behavioral control, all of them driving the intentions to behave ecologically simultaneously [3, 33]. When looking at the literature on consumers' willingness and acceptance of different circular business including PaaS models, such as renting, leasing, or sharing, circular economy, and green models, for electrical and electronic equipment, through the lens of TPB, a search conducted on the Scopus database identified 157 articles as shown in Table 1. After reviewing the abstracts, only 30 articles were found relevant, and four areas of scientific inquiry that have dominated research agendas were identified by the authors: (a) green appliances and technologies adoption with 47%, (b) any green product acceptance with 37%, EEE related products with 30%, (c) remanufactured products willingness to accept with 10%, and e-waste reverse logistics and recycling willingness with 6%.

For the first category, the revised studies analyze highly energy efficient appliances acquisition or eco-friendly products through the TPB framework and the use of some extensions such as, quality, environmental concern, eco-labeling, perceived value, and innovation diffusion, are often emphasized, particularly in developing economies (Pakistan, China, Malaysia, South Africa, and Indonesia) where decision-making dynamics differ from devel-

Table 1 *Search terms using Scopus

Boolean Operators	Search String	Categories	Quantity
	1st Search	"Product as a Service" OR "PaaS" OR "circular business" OR "circular economy" OR "rent*" OR "leas*" OR "product-service" OR "sustainable model" OR "subscription" OR "shar*"	4,440,299
AND	2nd Search	"acceptance" OR "willingness" OR "adopt*"	195,613
AND	3rd Search	"Theory of planned behavior*" OR "TPB"	920
AND	4th Search	"electric*" OR "electronic*" OR "appliance" OR "good" OR "EE"	157

oped regions [4, 5, 10, 20, 56, 72, 74, 98]. For the second category, found studies explore the determinants of sustainable and circular purchasing intentions using TPB framework and extending it with other constructs such as environmental awareness and perceived value, product involvement and altruistic motivations [31, [59, 63]. Studies in the third category focused on comprehending different consumer behaviors and their willingness to purchase remanufactured equipment using the TPB framework and highlighting the roles of functional value, perceived product quality, and price advantages as key extension constructs [47, 76, 77]. For the final fourth cluster, the two reviewed papers converge on consumer behavior and decision-making process concerning e-waste recycling, underscoring the importance of habits, perceived usefulness, and ease of recycling, extending the TPB framework in shaping pro-environmental actions. All studies predominantly rely on structured surveys with sample sizes ranging from 102 to over 485 respondents, analyzed using partial least square structural equation modeling (PLS-SEM) [1, 50].

Ecuador, as a developing South American economy, presents a unique opportunity for analyzing the drivers towards leasing [44]. Firstly, Ecuador presents a significant case study, ranking number eight in terms of population size, with approximately 17 million inhabitants, for Latin America. Due to its lack of surface, in comparison with other countries of the region, Ecuador emerges in first place in terms of population density for the South American region (World-Bank [89]). Secondly, Ecuador presents a unique case among other Latin nations, given that its currency is American dollars, a characteristic shared with Panama and El Salvador only [87]. Thirdly and lastly, the exploration of circular business models, particularly leasing, is still an emerging field. Even though previous research exists on green behaviors in Ecuador [40, 44, 54, 61], none of them have specifically explored leasing intentions for household appliances.

As shown with the four identified clusters, expanding the TPB model by integrating additional constructs, can improve the explanation power [86]. Environmental consciousness has been used by many scholars in the four clusters [8, 72, 98], and they agreed that individuals' feelings and judgement of the consequences of their actions on environmental problems are key factors in the adoption of green products and sustainable business models. Research indicates that consumers from developed nations generally exhibit greater environmental awareness than their counterparts in developing countries [31]. In this context, external influences such as national regulations, societal norms, collective agreements and stakeholder expectations diffused through mass media and social platforms, can shape individuals' awareness, perception and emotions on different environmental issues [40].

The present study adds valuable insights to the body of literature, particularly within the context of PaaS acceptance in developing South-American countries. Over the past eight years, studies have been predominantly originated on the preference of high energy performant appliances and green products from developing nations in Asia and Africa [4, 5, 10, 20, 31, 56, 59, 63, 72, 74, 77, 98]. In contrast, research focused on consumers' PaaS leasing intentions in developing South-American economies, such as Ecuador, remains scarce. Building on this, the proposed framework integrates the fundamental constructs of the theory of planned behavior with extended elements like environmental consciousness and external influences, as suggested in earlier works (e.g., Zhang et al., [8, 40, 72, 98]. This research is among the first to evaluate environmental concern in a moderation factor and external influences working as a mediator between consumers' intentions to behave ecologically and consumers leasing intentions of home appliances. The research framework

is outlined in Sect. 2. Section 3 presents the results assessing the validity and reliability through confirmatory factor analysis and structural equation modeling. Section 4 discusses the findings, addressing implications and limitations, and finally, Sect. 6 concludes by summarizing the key findings of the study.

Research Framework

In both developed and emerging markets, home appliances represent a growing substantial segment of consumer expenditures tied to population growth and welfare improvements, leading to an increase in raw materials scarcity, energy consumption and waste generation. Through the adoption of circular business models, especially leasing PaaS model, consumers can access appliances without owning them, which promotes longer product lifespans, refurbishment, and reuse. This approach not only curtails the demand for new appliances but also significantly diminishes their associated environmental footprint [17]. One key challenge for their development is consumers' concerns about the lack of ownership, which significantly reduces consumer interest in these sorts of business model structures [15]. Comprehending user expectations, desires, and motivations in PaaS systems presents a significant challenge for OEMs, which must be addressed during the circular business model development (Fig. 1).

Except for a few secondary findings [28, 44], nothing is known about how psychographic antecedents like attitudes affect consumers' decisions to choose leasing home appliances. A more profound understanding of users' mentality through an extended theory of planned

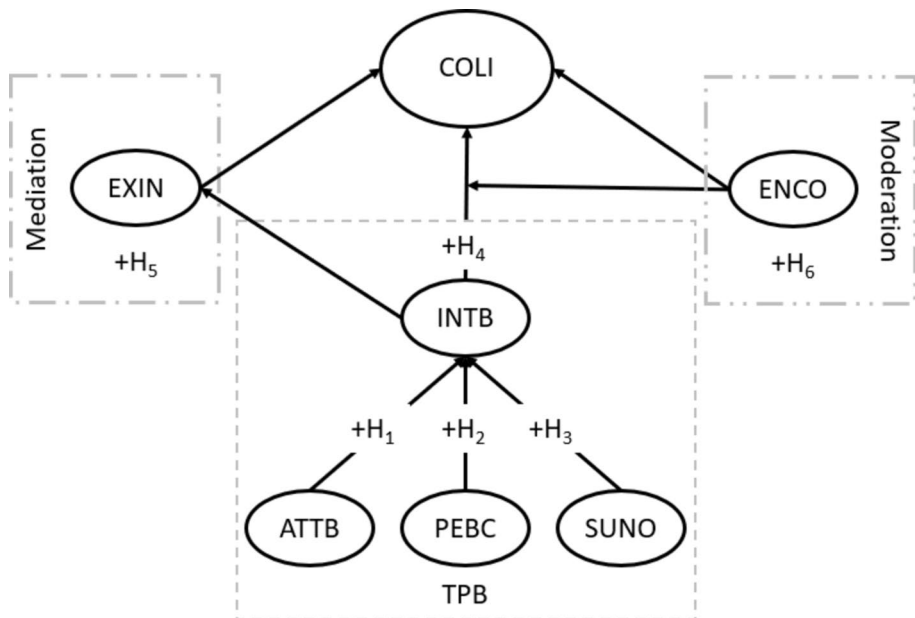


Fig. 1 Proposed research framework. Note ATTB: Attitude Toward Behavior; COLI: Consumer's Leasing Intentions; ENCO: Environmental Consciousness; EXIN: External Influences; INTB: Intentions to Behave Ecologically; SUNO: Subjective Norms; PEBC: Perceived Behavioral Control

behavior, can facilitate the creation of innovative PaaS leasing models that incorporates value alongside cognitive and experiential benefits, thus adopting a user-centric approach [42].

The Theory of Planned Behavior

The theory of planned behavior (TPB) was selected for this study as it offers a robust framework for understanding consumer behavior through three psychological constructs: attitude toward behavior (ATTB), subjective norms (SUNO), and perceived behavioral control (PEBC) [3, 28, 30].

TPB facilitates quantitative measurement of consumer intentions and influencing factors, providing empirical support for hypotheses validation around consumer behavior in a circular economy [15]. It allows researchers to examine the effects of individual traits, social context, and non-volitional factors on behavioral intentions. Moreover, TPB posits that individuals must feel capable of performing an environmentally friendly action to engage on it [38].

Within the framework of TPB, a person's reflection of their attitudes towards a specific subject as whether favorable or unfavorable is achieved. This influences their propensity to participate in different pro-environmental behaviors, such as waste sorting, separation and recycling [18, 30]. Individuals' relationships with family members, colleagues, close friends and even from social media and the news can influence their subjective norms, so affecting them can become a social pressure moving the individuals to support a particular circular behavior [80]. Additionally, according to the theory, PEBC predicts both intention and action by representing an individual's impression of having the necessary behavioral capabilities and opportunities [2].

This theory has proven especially valuable in the examination of consumers' intentions to choose circular business models [25, 84], as it proposes that the likeliness of individuals engagement in environmentally responsible activities is attached to their believe in attaining significant rewards, and/or if they hold in high regard to endorse the behavior, and/or if they perceive themselves as capable and adequately resourced to execute the behavior proficiently.

Attitude-intention-behavior models such as TPB frequently ignore the impact of external environmental and situational elements on customer purchase behavior [46]. To overcome these inadequacies, different ways to ameliorate TPB explanatory power and by doing so, improving the acceptance of CBMs, are the inclusion of other constructs working as adjustments to facilitate the different disparities between attitudes and behaviors found by previous researchers.

Extending this theory by including specific constructs relevant to circular business models, can help the framework to more accurately address the unique challenges and facilitators for the home appliances' market in these developing regions, such as overcoming the stigma of inferior quality or highlighting the environmental benefits [15, 28]. For this study, additional criteria such as environmental consciousness (ENCO) and external influences (EXIN) were discovered to be particularly useful for measuring customer approval of PaaS leasing business models.

ENCO can be described as a multidimensional concept focused on the inclination to participate in pro-environmental behaviors. When conceptualized in this way, environmental

consciousness aligns closely with the attitudinal or psychological dimension that underpins pro-environmental behavior [16]. This consciousness influences consumer behavior towards making more sustainable and circular consumption choices. Population with intrinsic higher senses of environmental consciousness are better prepared and more likely to adopt behaviors conducive to a circular economy, including recycling, extending the lifespan of products, and selecting goods that minimize environmental impact [57].

EXIN are defined as influences that stem from outside the individual, such as peers, marketing activities, and initiatives by governmental entities and other non-governmental ones (NGOs) [34]. These external influences appeal consumer behavior by molding their environmental knowledge and worries, which then influence their green purchase intentions. External stimuli to the consumer such as advertisements, branding, peer opinions, and institutional initiatives, significantly influence how consumers process information and decide on whether to adopt or not environmentally friendly products [51].

Intention To Behave Ecologically

A variety of factors influence behavioral intentions, which reflect an individual's commitment to taking specified activities [29]. They signify the extent of motivation and effort individuals are prepared to invest in carrying out a specific practice. As proposed by Ajzen [3], stronger intentions serve as an indicator of an increased probability of actually implementing the behavior. In the realm of environmental psychology, exploring the correlation between environmentally conscious behavioral intentions and subsequent eco-friendly actions, such as that the desire in engaging in ecological conduct can predict total ecological behavior (e.g., the intention to buy bio-organic items and actually executing these actions), can hold considerable significance [49]. Drawing upon these findings, the following hypothesis is put forward:

H1. Intention to behave ecologically is positively related to consumers' leasing intentions of home appliances.

Attitude Toward Behavior

The attitude toward a given activity or behavioral goal summarizes their overall judgment of that conduct. This encompasses the awareness, beliefs, and both favorable and unfavorable predispositions arriving to individuals when contemplating the behavior [3]. Essentially, it determines whether the individual perceives the behavior positively or negatively. Having a positive attitude toward behaviors supporting different environmental concerns is likely to motivate people to conduct ecologically beneficial acts or make more sustainable decisions. Several previous research has identified the positive relationship between attitude, costumer mentality, consumer sentiments, and behavioral intention on green product purchase across different cultural contexts [68, 95]. Consequently, we present the following hypothesis:

H2. Attitude toward leasing behavior is positively related to the intentions to behave ecologically.

Perceived Behavioral Control

Perceived Behavioral Control (PEBC) involves a person's assessment of the level of easiness or difficulty a specific action may take finally impacting their willingness to execute them [3]. High PEBC means the individuals sense acting of specific behaviors is relatively easy, while on the other hand, low PEBC relates to individuals finding an action very challenging. This concept faces both self-confidence and self-efficacy, one against the other, and they are influenced by internal factors like skills and external factors such as resources. Different empirical research highlights PEBC's importance in predicting behaviors and intentions, especially for actions not entirely within one's control, such as purchasing eco-friendly products [9, 58, 68];). Based on these insights, we propose the following hypothesis:

H3. Perceived behavioral control is positively related to the intentions to behave ecologically.

Subjective Norms

Subjective Norms (SUNO), as defined by Ajzen [3] cited in Han et al., [38], refer to the influence that significant others have on an someone's deportments and ideologies. This construct reflects the social pressure an individual feels to perform or avoid a specific action, such as adopting green business models. When consumers perceive that their peers are inclined to adopt green business models, they are likely to develop similar intentions [24, 55, 62, 90]. Thus, we propose the following hypothesis:

H4. Subjective Norm is positively related to the intentions to behave ecologically.

The Mediating Role of External Influences

Mediation is a research approach wherein an intermediary variable, called the mediator, facilitates the relationship between two related phenomena. Researchers employ this analytical approach to determine if modifications in the independent construct cause changes in the mediator variable, which in turn affect the dependent construct within the model [27].

External pressures and internal ethical beliefs significantly shape ethical attitudes and decision-making [23]. External influences include social factors like situational conditions, significant events, and opinions from others, encompassing stakeholder demands, regulations, societal expectations, consensus, and perceived benefits or drawbacks. Engagements with these elements affect individuals' interpretations and viewpoints. Supported and approved societal behaviors are frequently mimicked by individuals to feel part of the community, making them susceptible to social influences. This study explores the potential mediating role of external influences in the relationship between the intention to behave ecologically and customers' intention to lease home appliances.

H5 External Influences mediate the correlation among Consumers' Leasing Intentions of Home Appliances and their Intention to Behave Ecologically.

The Moderating Effect of Environmental Consciousness

Moderation arises when a third party (variable), known as the moderator, affects the relationship between two connected constructs. This moderator can affect the strength of their association and even change the sense of direction (positive/negative), meaning that the interaction between the two constructs fluctuates based on the moderator's power. Environmental consciousness (ENCO), known as environmental awareness sometimes, encompasses individuals' perception on the environmental problem and how this perception affects other psychological constructs that motivate individuals to participate in pro-environmental activities [96]. According to Newhouse [67], attitudes and emotions toward others or objects play a crucial role in shaping human behavior. In this research, environmental consciousness acts as a mediator, bridging the relationship between customers' intentions to adopt eco-friendly behaviors and their willingness to lease home appliances.

H6 Environmental Consciousness positively moderates the positive relationship between intentions to behave ecologically and consumers' intentions to lease home appliances.

Survey Design and Implementation

The survey consisted of two components: The first section demanded information from all seven construct items identified in the previous sub-sections. Three items were used to measure Attitude Toward the Behavior (ATTB), Subjective Norms (SUNO), and Perceived Behavioral Control (PEBC). Four items were used to assess Intention to Behave (INTB) (refer to supplementary material for questions) [78, 81, 94]; Fishbein and Ajzen, 1977). Consumer Leasing Intentions (COLI) and Environmental Consciousness (ENCO) were each measured using five items [71, 88, 93, 97]. Lastly, External Influences (EXIN) were evaluated using five items derived from two researchers [19, 82]. The second section of the survey demanded socio-demographic information of both the household head and their family members. Drawing from prior research conducted in the corresponding field, the questionnaire looked for data related to the gender, age, educational attainment of the head of the household, along with general family details including household size and family income [43], 2022). Through the administration of twenty pre-tests, the survey was validated, allowing the researchers to correct mistakes and rephrase any crucial queries and delete some of them if disparities were discovered, as was also done by Hidalgo-Crespo et al. [40, 44].

The suitable population for this study consisted of household heads aged 18 and older who hold decision-making authority regarding the adoption of specific circular business models for their families [14, 41]. The respondents were offered a 7-point Likert scale for all 28 measurement items going from 1 (totally disagree) to 7 (totally agree). The 7-point Likert scale offers a highly reliable, simple, and valid technique to measure different psychological constructs with precision. It permits finer distinctions in responses compared to a 5-point scale, allowing for greater variability in the responses and enhancing the bias between respondents' perceptions or opinions. Additionally, participants find 7-point scale more intuitive and natural compared to other formats [39, 69].

Data were collected through personal interviews and written questionnaires from January to March 2023 (see Supplementary Information). Face-to-face interviews provided

respondents with sufficient time to thoughtfully consider their responses, and interviewers were available to address any queries or concerns [73]. The data collection involved 120 students from two local universities in Guayaquil City. They approached family heads in their neighborhoods or other family members who were not residing with them. Because numerous students were enrolled in online courses and resided beyond the city limits, this study achieved extensive geographic coverage, encompassing a considerable number of participants from various regions in Ecuador in an economically efficient manner [99].

Sample Size and Composition

The sample size for this investigation was determined based on the guidelines provided by Hair et al. [37], which recommend obtaining 15–20 observations per variable analyzed. This criterion was also employed by Hidalgo-Crespo and Amaya-Rivas [40]. The study investigated seven constructs and 28 items. Based on this, an appropriate sample size would be 560 respondents (28 variables $\times$ 20 observations per variable). Nevertheless, the research employed a substantial sample size of 1,026 participants, well exceeding the suggested minimum of 385 for infinite populations as recommended by Kwak and Kim [52]. This ensured a high level of confidence at 95%, with a narrow margin of error of 3% for infinite populations. Table 2's descriptive statistics indicate that the sample was primarily composed of married males aged 37 to 47 years, who possess a bachelor's degree, belong to families of four, and have an annual income ranging from \$451 to \$900 USD.

Table 2 Sample characteristics

Variable	Categories	Frequency	Percentage*
Gender	Male	549	53.51
	Female	477	46.49
Family Size	1–2	138	13.45
	3	232	22.61
	4	322	31.38
	5	221	21.54
	6	80	7.80
	>6	33	3.22
Age of household head	18–25	80	7.80
	26–36	242	23.59
	37–47	305	29.73
	48–58	260	25.34
	59–69	115	11.21
	>69	24	2.34
Education of household head	None	19	1.85
	Primary School	90	8.77
	High School	419	40.84
	Bachelor	430	41.91
	Post-Graduate	68	6.63
Family Income USD	<450	195	19.01
	451–900	395	38.50
	901–1,350	200	19.43
	1,351–1,800	116	11.31
	1,801–2,250	102	9.94
	>2,250	18	1.75

Remark: *Calculations made dividing by a total sample of 1,026 respondents.

Results

For this research, SmartPLS 4.2 and partial least squares structural equation modeling (PLS-SEM) were employed for data analysis. Previous studies have shown PLS-SEM to be reliable in environmental psychology research [26, 70].

Measurement Model

Convergent and discriminant validity of the indicators was examined prior to assessing the constructs validity and reliability. Each construct showed factor loadings ≥ 0.5 and Cronbach's alpha > 0.7 (Table 3), indicating high internal consistency and reliability. Average variance extracted (AVE) for all constructs exceeded 0.5, demonstrating strong convergent validity. The results confirm strong convergent validity across all constructs, with indicators effectively capturing the essence of their respective latent variables.

Discriminant validity was confirmed by comparing square roots of AVE values to inter-construct correlations in Table 4, meeting required criteria. This indicates that each construct is distinct and does not excessively overlap with others.

Table 3 Convergent validity

Attribute	Component	External loading	Cronbach's Alpha	AVE	Consistency Reliability
Attitude toward behavior (TPB)	ATTB1	0.688	0.806	0.582	0.814
	ATTB2	0.744			
	ATTB3	0.849			
Perceived Behavioral Control (PEBC)	PEBC1	0.683	0.757	0.510	0.759
	PEBC2	0.714			
	PEBC3	0.745			
Subjective Norms (SUNO)	SUNO1	0.788	0.763	0.523	0.781
	SUNO2	0.570			
	SUNO3	0.790			
Intentions to Behave Ecologically (INTB)	INTB1	0.763	0.824	0.541	0.829
	INTB2	0.714			
	INTB3	0.798			
	INTB4	0.661			
Consumers' Leasing Intentions (COLI)	COLI1	0.754	0.859	0.550	0.859
	COLI2	0.730			
	COLI3	0.762			
	COLI4	0.737			
	COLI5	0.723			
Environmental Consciousness (ENCO)	ENCO1	X	0.796	0.590	0.865
	ENCO2	0.658			
	ENCO3	0.593			
	ENCO4	0.993			
	ENCO5	X			
External Influences (EXIN)	EXIN1	0.690	0.836	0.505	0.836
	EXIN2	0.674			
	EXIN3	0.719			
	EXIN4	0.737			
	EXIN5	0.729			

Table 4 Discriminant validity

Constructs	ATTB	ENCO	EXIN	INTB	PEBC	COLI	SUNO
ATTB	0.763						
ENCO	0.015	0.768					
EXIN	0.690	0.038	0.710				
INTB	0.738	−0.067	0.661	0.736			
PEBC	0.705	0.182	0.600	0.675	0.714		
COLI	0.719	0.111	0.661	0.734	0.615	0.741	
SUNO	0.722	−0.010	0.700	0.720	0.707	0.672	0.723

Table 5 Hypotheses validation

Hypotheses	β	SD	t-value	p-value	Results
H1: ATTB $\rightarrow$ INTB	0.492	0.107	4.586	0.000*	Supported
H2: PEBC $\rightarrow$ INTB	0.027	0.079	0.344	0.731	Not Supported
H3: SUNO $\rightarrow$ INTB	0.347	0.081	4.299	0.000*	Supported
H4: INTB $\rightarrow$ COLI	0.498	0.102	4.379	0.000*	Supported

P-value probability value, *t*-value *t*-statistics, *SD* Standard Deviation, β Beta Coefficient. Relationships are significant at

* $p < 0.001$

Structural Model Assessment

After evaluating the measurement model, the subsequent step involves estimating the structural path to assess the path coefficients. These coefficients analyze the relationships between the study's constructs and determine their statistical significance. The results are presented in Table 5; Fig. 2.

Hypothesis 1 (H_1) investigates whether attitude toward behavior significantly and positively influences the intention to engage in environmentally friendly actions. The results reveal that ATTB has a favorable and noteworthy effect on the INTB ($\beta = 0.492$, $p < 0.001$). Thus, H_1 was supported. The strong positive effect of ATTB on INTB to engage in eco-friendly actions reaffirms the role of individual evaluation as a precursor to intention, emphasizing the criticality of fostering favorable attitudes.

Hypothesis 2 (H_2) explores whether a significant and positive relationship exists between perceived behavioral control and the intention to engage in environmentally friendly behavior. The results reveal that PEBC has an unworthy effect on the INTB ($\beta = 0.027$, $p > 0.05$). Consequently, H_2 is not supported. This result is unexpected, diverging from established TPB research. This might indicate that contextual factors such as infrastructural or socioeconomic barriers in leasing appliances do not weaken consumers' intentions to adopt leasing models.

Hypothesis 3 (H_3) evaluates whether subjective norms significantly and positively influence the intention to engage in environmentally friendly behavior. The results reveal that SUNO has a favorable and noteworthy effect on the INTB ($\beta = 0.347$, $p < 0.001$). Thus, H_3 is supported. This significant influence appoints that consumers' decision making are indeed affected by social expectations and peer behavior.

Hypothesis 4 (H_4) investigates whether there is a significant and positive relationship between the intention to engage in environmentally friendly behavior and consumers'

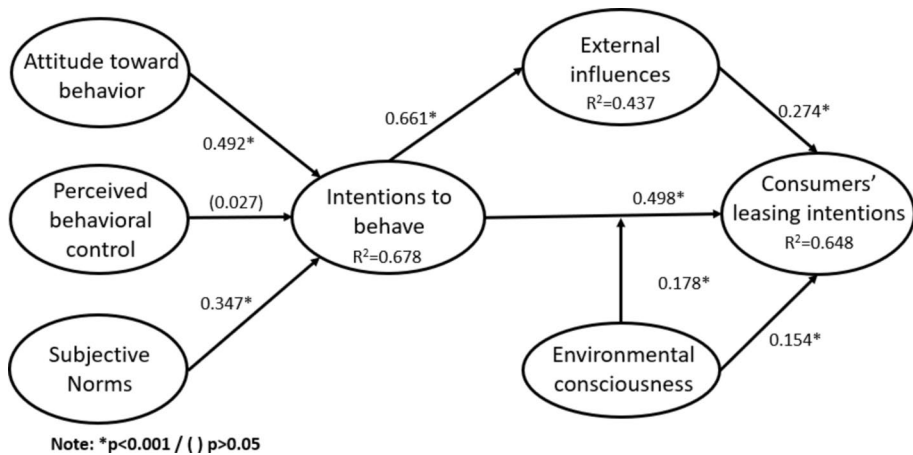


Fig. 2 Extended TPB structural model. Note: Numbers correspond to the coefficients (β values). Those coefficients with an * mean that the relationship is accepted with a probability value of less than 0.001. The coefficient between parentheses means that there is not significant relationship due to a probability value higher than 0.05

Table 6 External influences mediating effect

Total Effects (INTB → COLI)	Direct Effect (INTB → COLI)	Indirect Effects of Intentions to behave on Consumers' leasing intention			
Coefficient	Coefficient	Hypothesis	Coefficient	Percentile bootstrap 95% confidence interval	
				Lower	Upper
0.679*	0.498*	H5: INTB → EXIN → COLI	0.181*	0.114	0.258

INTB Intentions to behave, COLI Consumers' leasing intentions, EXIN External influences

* p -value < 0.001

intention to lease. The results demonstrate a substantial and favorable effect of the INTB on COLI ($\beta=0.498$, $p<0.001$). Hence, **H₄** is upheld. The strong positive relationship reinforces the direct connection between intention and behavior solidifying the applicability of TPB in predicting actual sustainable consumption behaviors.

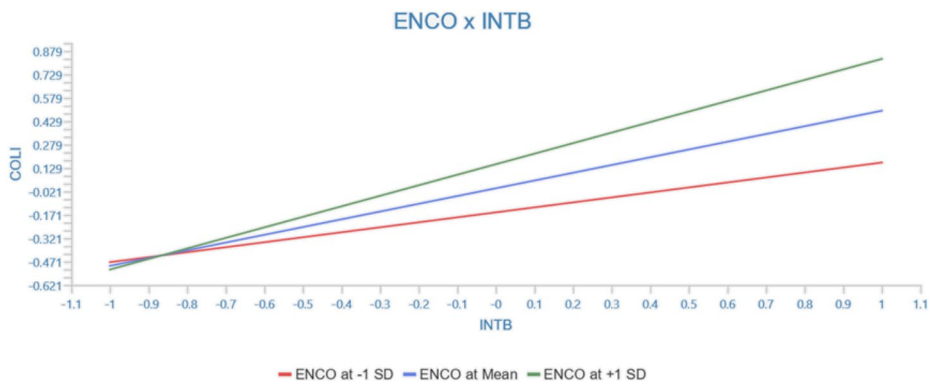
Mediation Analysis

The results of the mediating effect of EXIN on the connection between INTB and COLI are detailed in Table 6 and showed that INTB has a statistically significant indirect effect on COLI through EXIN (Hypothesis 5: $\beta=0.181$, $p<0.001$). Furthermore, the total effect of INTB on COLI was also significant ($\beta=0.679$, $p<0.001$). Including the mediator in the model, the direct effect of INTB on COLI remained statistically significant ($\beta=0.661$, $p<0.001$), thus affirming Hypothesis 5. These findings indicate that EXIN serves as a partial mediator, enhancing the relationship between INTB and COLI.

Table 7 Environmental consciousness moderation analysis

Relationship	Coefficient	SD
Moderating Effect (ENCO*INTB) -> COLI	0.178*	0.007
ENCO -> COLI	0.154*	0.015
INTB -> COLI	0.498*	0.107

ENCO Environmental Consciousness, COLI Consumers' leasing intentions, INTB Intentions to behave
* p -value < 0.001

**Fig. 3** Moderator slope analysis. Note ENCO: Environmental Consciousness; COLI: Consumer's Leasing Intentions; INTB: Intentions to Behave

The significant indirect effect of INTB on COLI suggests that EXIN plays a role in enhancing consumers' leasing intentions, however, they do not fully account for the relationship, due to the strong direct effect of INTB on COLI.

Moderation Analysis

The effect to moderate of Environmental Consciousness (ENCO) is analyzed in Table 7. Initially, the R-squared value was 0.605, explaining 60.5% of the variance in COLI by INTB and EXIN. Incorporating ENCO increased the R-squared value to 64.8%, a 4.3% improvement. The analysis revealed that ENCO positively and significantly influences the INTB-COLI relationship ($\beta=0.178$, $p<0.001$), supporting Hypothesis 6. Additionally, a slope analysis (Fig. 3) was employed to further elucidate the moderating effects. The graph shows that the line steepens with higher levels of ENCO, indicating that INTB has a greater impact on COLI in these cases than at lower ENCO levels.

However, at lower ENCO levels, the line appears to flatten, suggesting that an increase in INTB may not lead to a proportional change in COLI. The F^2 effect size was calculated to be 0.073. According to Cohen's (1988) guidelines, values of 0.02, 0.15, and 0.35 represent small, medium, and large effects, respectively. This indicates that the moderating effect plays a significant role in explaining the variance in the endogenous construct (COLI). The increase in R-squared value (from 60.5 to 64.8%) together with the slope analysis confirm that ENCO strengthens the INTB-COLI link, with the effect being more pronounced with higher ENCO levels. Higher ENCO can be correlated with increased pro-environmental

Table 8 Predictive capability

Predictor(s)	Outcome(s)	R^2	F^2	Q^2
INTB	COLI	0.648	0.233	0.399
ENCO			0.031	
EXIN			0.112	
ENCOxINTB			0.046	
INTB	EXIN	0.437	0.435	0.356
ATTB	INTB	0.678	0.134	0.498
PEBC			0.018	
SUNO			0.097	

values and willingness to adopt circular economy solutions like leasing, suggesting that fostering ENCO among consumers with high recognition of environmental problems can lead to higher adoption rates, while low ENCO level population should be targeted with tailored interventions.

Explanatory Power

The R^2 statistic measures the proportion of variance in the dependent variable explained by the independent variables, reflecting the model's explanatory power and in-sample predictive ability [32]. Higher R^2 values indicate stronger explanatory strength, with Cohen [21] setting benchmarks at 0.26 (substantial), 0.13 (moderate), and 0.02 (weak). Acceptable R^2 values can vary by context; for example, 0.10 may suffice in stock return prediction [22]. In this study, all endogenous constructs had R^2 values above 0.44, demonstrating strong explanatory power. Effect size (F^2) assesses the impact of removing an exogenous construct, with Cohen defining 0.35 as substantial, 0.15 as moderate, and 0.02 as small. Table 8 shows F^2 values for PEBC on INTB from 0.018 to 0.435, and Q^2 values for all endogenous constructs above zero, confirming predictive relevance.

The strong R^2 (0.678) and F^2 (0.134) values suggest that ATTB is a major contributor to INTB. This can lead to a higher likelihood of individuals forming ecological intentions when attitudes are favorable. Moderate F^2 (0.067) value of SUNO on INTB indicates that while subjective norms are influential, they act alongside other factors like attitudes and external influences. The strong R^2 (0.648) and F^2 (0.233) indicate that INTB is a primary driver of leasing behavior, suggesting that efforts to enhance intentions directly impact adoption. The model's high predictability ensures its applicability in real-world contexts, enabling different actors to anticipate consumer behavior effectively.

Discussion

The findings of this research indicate that extending the theory of planned behavior with external influences and environmental consciousness in both mediator and moderator roles improves the robustness of the TPB model for predicting consumers' leasing intentions of home appliances in Ecuador.

The strong found correlation between attitude and leasing intentions align with prior studies exploring remanufactured products and other circular business models. This emphasizes that consumer attitudes, especially towards product quality, risks, and price advan-

tages, play a critical role in adoption decisions [47, 76]. Similarly, Scheller et al. [72] noted that attitudes drive the subsequent adoption intentions for low-carbon technologies like energy-efficient appliances and Chi et al. [20], identified attitudes as a primary drive for sustainable intentions.

The significant influence of subjective norms highlights the critical role of social and cultural influences and family and friends perceived approval in shaping consumers intentions for green behaviors. This resonates with other findings which demonstrated that social and normative pressures play a pivotal role in encouraging green product and business model choices [50, 59, 76, 77].

The non-validation of perceived behavioral control suggests that even when individuals have favorable attitudes and are influenced by social norms, perceived barriers to action may impede their intentions, aligning with other studies that mentioned the reduction of PEBC for e-waste recycling or other behavioral intentions when structural limitations and logistical challenges are found [50, 65].

The strong positive relationship between the intentions to behave and consumers' leasing intentions reaffirms TPB's framework emphasizing that individuals with high ecological intentions are likely to perform actionable behaviors. This strong link further validates previous research on circular economy initiatives, which found that pro-environmental intentions strongly affect the adoption of circular business models [28, 79]. Similarly, the intentions to recycle e-waste were strongly linked to actual recycling behaviors, highlighting the translation of intentions into measurable behaviors [50]. Finally, just as price adjustments impact in commodity markets, different triggers that could impact leasing decisions should be analyzed [91].

The results additionally indicate that external influences (EXIN) partially mediate the relationship between intentions to behave ecologically (INTB) and consumers' leasing intentions (COLI). This suggests that mass media, internet access, and influencer power play a critical role in reinforcing and facilitating the transition from intention to action. The partial mediation indicates that external influences can further improve the model, but must be complemented by other internal factors, such as strong environmental attitudes and values. This aligns with other findings such as the influence of government policies and social norms in e-waste recycling [50], the relationship between peer influence and societal expectations in the acceptance of remanufactured products [76], the noted adoption of circular business models driven by government incentives and social endorsement [28], and how housing price dynamics and policy interventions influence Chinese household demand [92].

The moderation analysis revealed that environmental consciousness significantly deepens the correlation between intentions to behave ecologically and consumers' leasing intentions. This suggests that environmental consciousness serves as an amplifier of the intention-behavior link aligning consumers' environmental personal values, sensitivity, and knowledge with sustainable consumption models. Previous studies have also highlighted that high environmentally conscious consumer are more motivated to engage in green purchasing behaviors and circular economy practices [28, 31, 63, 68–75]. Understanding the limited recovery of critical elements in e-waste and their increasing global scarcity, can reflect on individuals' urgency to adopt circular solutions [6].

The high level of collectivism that characterizes the South American region places significant ties, community expectations, and group identity in the decision making. In such context, the high levels of subjective norms influencing the intentions to behave ecologically

found in this research, suggests the further important role of family, friends, and society in general compared to personal attitudes and self-determination when making decisions. This is in line with previous literature, where collectivist cultural settings increased the role of subjective norms in encouraging circular practices and green products adoption [50, 77].

This study has demonstrated the extended TPB model's effectiveness as a suitable research framework for understanding consumers' leasing intentions. Furthermore, this study contributes to addressing the persistent intention-behavior gap normally observed in sustainable consumption literature. While TPB framework states that consumers showing strong intentions to act sustainability, other research indicates that external barriers can weaken the realization of these intentions [76]. However, this study brings light to emerging markets like Ecuador, suggesting that leasing business model may offer an accessible and feasible solution, where resource limitations and affordability constraints are prevalent.

Theoretical Implications

The primary apport of this study is the discovery of a significant and positive relationship between the intention to engage in environmentally friendly acts and consumers' intents to lease home appliances. This highlights the power of the TPB model in capturing actual behavior, suggesting that these intentions need to be fortified through the improvement of their specific drivers. The strong impact of constructs such as attitude towards behavior and subjective norms on the intentions to behave ecologically highlight how both internal drivers (positive sensations) and external drivers (social approval) can work synergistically to motivate behavior. While attitude towards behavior reflects the importance of individuals' cognitive evaluations regarding environmentally-friendly outcomes (e.g., waste reduction, energy efficiency), subjective norms reflect on individuals' perception of the value of collective approval or disapproval, making them feel comfortable around socially endorsed behaviors like leasing home appliances.

Interestingly, perceived behavioral control, although considered a good predictor for green business model acceptance in previous studies, did not show a significant relationship with consumers' leasing intentions in this research. Consumers feel that the "perceived ease or difficulty of a behavior" is not that important a factor for following green business models. One possible explanation could be that consumers do not perceive significant obstacles or complexities in adopting leasing models due to the expectation of high-quality systems such as service platforms and costumer attention before, during and after contract signature, that they expect to simplify the process for consumers. Another possible explanation is the lack of sufficient experience or familiarity of the surveyed population with relatively new leasing models in an emerging economy like Ecuador. Therefore, the theoretical implication of perceived behavioral control in green behavioral intentions may be contingent on the familiarity and maturity of the business models being assessed.

This study improves the understanding of the role of environmental consciousness as a significant moderator between consumers' intentions to engage in environmentally friendly behaviors and their propensity to lease home appliances, demonstrating how integrating personal ecological values enhances the explanatory power of sustainable behavioral models. The improvement of explanatory power in almost five points suggests that behavioral intentions alone may not be enough to predict actions unless they are reinforced by powerful

environmental consciousness, particularly in emerging markets where sustainable practices like leasing are still evolving.

Finally, the research reveals external influences as a partial mediator between ecological behavioral intentions and consumers' intentions to lease. This implies the broader societal, economic, and structural dynamics importance in shaping pro-environmental behaviors. Given the partial mediation, these external influences act as facilitators or reinforcers of internal motivations playing a complementary role in the reduction of the intention-behavior gap, by addressing perceived barriers, fostering trust, and normalizing leasing behaviors within society.

Practical Implications

The results of this paper can infer some trade-offs consumers are willing to make, giving both policymakers and original equipment manufacturers, sufficient information for tailored offerings to maximize leasing acceptance. Some strategies to improve positive attitudes could be educational campaigns about the environmental benefits and cost-saving advantages of leasing home appliances. Efficient advertising approaches should also be utilized to inform consumers about the benefits of leasing in comparison to buying, highlighting advantages such as maintenance agreements, after-sales service, and eco-friendly end-of-life disposal choices. Public endorsements or green certifications to those businesses considering changing their business model together with communication efforts could further enhance consumer confidence and foster their positive attitudes. Business could benefit from social marketing and endorsements (due to the high social norms influence) by respected community leaders trying to reach as much population as possible enhancing normative pressure. Different public campaigns to make leasing the socially responsible choice and OEMs association with NGOs and well-known and respected community associations could further strengthen social acceptance. Other ways to bridge the intention-behavior gap could be by addressing potential barriers of possible consumers during their decision-making offering free trials, reduced initial affiliation costs and the liberty to withdrawn contracts without penalties. Additionally, governmental efforts could consider tax reduction initiatives for both OEMs and consumers wanting to adopt leasing models. Digital and social network publicity with the help of well-known internet figures (influencers of different platforms) can further amplify leasing model attractiveness, especially for the younger internet-connected generation. On the matter of environmental consciousness, marketing efforts should be segmented based on different levels. For highly conscious consumers, campaigns could focus on tangible benefits other than environmental perks, such as the access to cost-saving energy-efficient high-quality appliances to further stimulate the appeal of leasing models. On the other hand, for low conscious consumers, campaigns highlighting the ecological benefits could help enhance the intention-behavior correlation.

Limitations and Future Studies

Although our investigation produced compelling findings, it is crucial to acknowledge various limitations. Specifically, reliance on cross-sectional self-report surveys instead of direct observations of real-life behaviors could potentially introduce biases such as response bias and faking good bias. These biases could conceivably lead to inaccurate reflections of the

situations, with risks of both under-reporting and over-reporting. Given that currently, in the country, there are no leasing schemes other than vehicles, we do not have real evidence of previous leasing behavior and we only count on the interest of such PaaS offerings during the explanation given to the population. Second, this study's convenience sampling method has produced a demographic bias toward Guayas province dominance, which deviates from the whole country, since cultural changes can be observed between the coastal zones and the highlands.

To improve the external validity and generalizability of the findings, it is advised that future research increase the data obtained from other geographical regions of the country to get a better picture of the whole distribution. Further, as various cultures and nations may yield various results in behavioral studies, a cross-cultural or cross-national study may be carried out to provide a better understanding of underlying issues in developing countries, maybe from Latin America. Future research could also cluster the results according to the different socio-demographic characteristics to be able to personalize different strategies according to the consumer segments. Finally, correlation, regression and variance analysis could further uncover other relationships missed by PLS-SEM.

Conclusions

Our results basically corroborate those of earlier extended TBP studies showing that a pro-sustainability mindset encourages the corresponding sustainable intention to follow sustainable business models when acquiring new home appliances. Building on this, our results indicate that consumers' intentions to behave ecologically are motivated mostly by their attitudes toward behavior, followed by their subjective norms. To strengthen the predictive potential of the leasing intention, this study goes beyond the previous ones by, on the one hand, considering only the leasing of home appliances through PaaS offering, and on the other hand, incorporating external influences and environmental consciousness as mediator and moderator, respectively.

For future research, it is recommended to replicate this study in other cities within the country, particularly those nearer to the highlands, to facilitate comparative analysis. Additionally, continuous validation of this model could be undertaken in other Hispanic developing nations. Ultimately, this psychological model could prove to be highly valuable in informing public decision-making.

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Data Availability Data will be made available on request.

Declarations

Competing Interests The authors declare no competing interests.

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